

READ ME FIRST · v1

LANA

Technical Architecture

the local agentic concierge · unified blueprint for the team

YUNCHAO · AI

ASJID · BACKEND

ABDULLAH · FRONTEND

2026-06-12 · founder authored · companion deck to LANA_BLUEPRINT_v1

WHAT THIS IS

17 slides · the technical narrative
Step 1, Step 2, Step 3 + 5 phases
Each slide ends with: reference docs
Footnotes ground us in 2026 research

USE THIS:

- 1 · read solo · 20 minutes
- 2 · come to Monday sync aligned
- 3 · keep docs open in parallel
- 4 · push back hard on what feels wrong

TL;DR • the thesis in one slide

what • why we are different • the moat

THE STACK

3

LAYERS of intent

Explicit • Out-of-scope • Latent

+ 6 NODE TYPES • 3 FLYWHEELS

heterogeneous knowledge graph

(people • places • things • activities • events • tips)

compounds with every captured signal,

identity claim, and feedback impression

LANA IS NOT A CHATBOT.

She is a local agentic concierge that compounds in value with every captured signal • every identity claim • every match made.

WE WIN BY COMBINING THREE LAYERS.

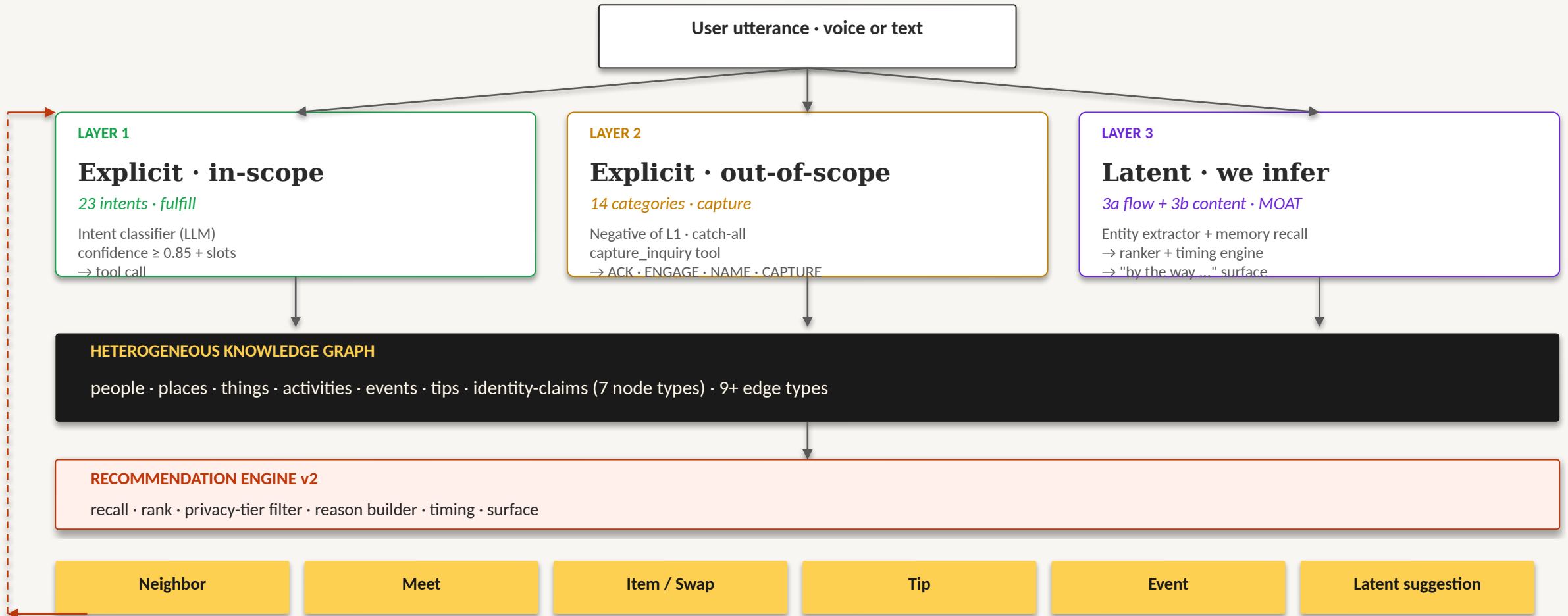
Explicit fulfillment (L1) • graceful out-of-scope capture (L2) • proactive latent inference (L3) — over a heterogeneous knowledge graph.

THE MOAT IS NOT THE LLM.

It is the per-block, per-mom, per-relationship data substrate that no general-purpose AI can replicate.

The unified Lana stack

every turn runs three parallel intent extractions • over one heterogeneous graph • feeding one ranker • closing one feedback loop



STEP 1

Layer 1 • Explicit Intent

what mom asks for, directly • the table-stakes classifier you already know

23 IN-SCOPE INTENTS • 7 CAPABILITY SURFACES

Discovery	find peers • find by attrs • block summary	3
Identity	add / edit claims • complete • show profile	4
Looking	swap-seek • meet-seek • tip-seek	3
Sharing	swap-offer • host-meet • tip-share	3
Relationship tier	send nudge • respond • advance ladder	5
Auth	signup / login / OTP / logout / photo	6
Settings / Help	name • ZIP • prefs • what-can-you-do	5

EXAMPLE TURN

"I'm looking for 3T rain boots"

- 1 Classifier (Sonnet/Hermes)**
intent: looking.swap • conf 0.92
- 2 Confidence ≥ 0.85 ?**
YES • all required slots filled
- 3 Tool call**
extract_looking_intent({item:'3T rain boots', ...})
- 4 Save signal**
local_signals row written
- 5 Match against block**
→ peer_matches returned to FE

STEP 2

Layer 2 · Out-of-Scope · roadmap-from-data

what mom asks that we don't fulfill · but capture · graceful refusal IS data collection

THE 4-STEP PATTERN (every out-of-scope turn)

ACK the underlying need (not the literal ask)
"That's a big move — moving is a lot. What's the trigger?"

ENGAGE 1-2 follow-up turns · texture the signal
"Closer to the school zone, or somewhere quieter?"

NAME the constraint warmly (6-12 words)
"I don't broker rentals yet — but a few moms have asked."

CAPTURE the signal explicitly with consent
"I'll keep this one. Want me to ping you if we line something up?"

BY THE NUMBERS

30-40%

of v0.1 traffic expected to be out-of-scope

14 expected categories · self-extending
Signal Aggregator clusters semantically
Roadmap CANDIDATES auto-detected

★ THE IP

*This is not a fallback.
This is the per-block roadmap-from-data flywheel.
Every refusal is a labeled, geocoded, time-stamped
demand signal · captured WITH CONSENT.*

*No competitor structures their refusal pattern
this way · it requires intent at architecture-time.*

Layer 3 · Latent Intent · the concierge layer

what mom didn't ask, but we inferred · the difference between a chatbot and a concierge

3a · FLOW-DRIVEN

Circumstantial

state machine + accumulated context

- mid-FTUE · value delivered + no phone yet · trigger SIGNUP CHIME-IN
- 7 days since last session · trigger CHECK-IN re-engagement
- new fellow accepted nudge · trigger NEXT-INTERACTION suggestion
- 4th identity claim added · trigger COMPLETE PROFILE soft-nudge

FRAMEWORKS

XState · LangGraph conditional edges · Sierra step-tracker

3b · CONTENT-DRIVEN

Inferential

entity extractor + capability index + ranker

mom: "my kid does karate"

→ *entity:karate(activity) extracted*

→ *capability matches: looking.meet (0.85) · looking.swap (0.78) · looking.tip (0.71)*

→ *ranker picks top 1-3 · timing engine queues*

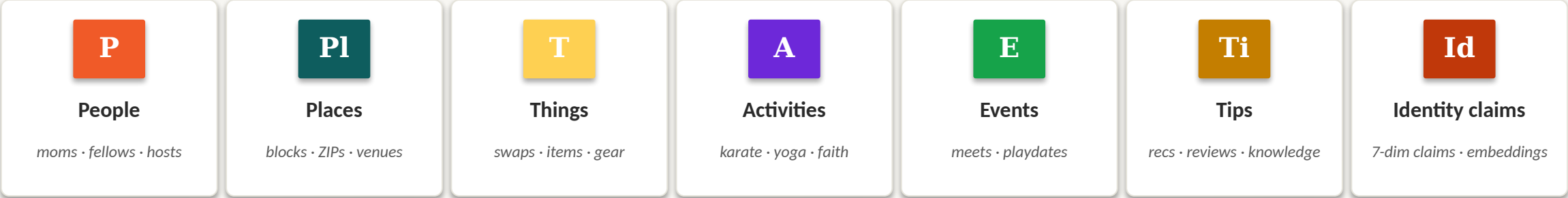
→ *Lana surfaces: "May I, Maria — moms in your block with karate kids meet Saturdays."*

★ IP · THIS IS WHAT HERMES 4, CLAUDE, OR GPT ALONE CANNOT DO

Layer 3 is the architecture ON TOP of the LLM. It requires: per-mom memory (Letta-style) + entity extractor (separate from intent classifier) + embedded capability index + timing engine + frequency caps. 6 months of architecture · 12 months of data collection to train · per-block specificity that only emerges from real usage. The first competitor to ship this wins the local-agentic-AI category.

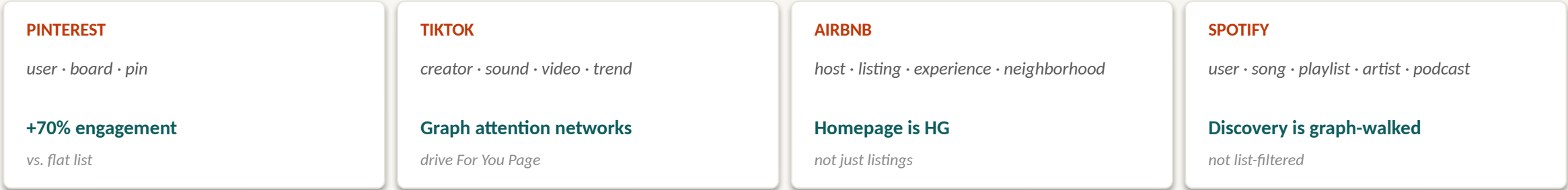
The data substrate · heterogeneous knowledge graph

Yunchao's v1 diagram treats this as one edge type (person → person). Reality is multi-typed.



9+ EDGE TYPES · proximity · trust tier · co-attendance · offers · seeks · hosts · attends · shares · references

WHY HETEROGENEOUS WINS · proven pattern from 2026's biggest recommendation systems



Recommendation Engine v2

Yunchao's v1 is correct as a peer-to-peer matchmaker · v2 makes it heterogeneous · multi-triggered · time-aware · bidirectionally-trusted

YUNCHAO'S v1 · KEEP

Peer-to-peer recommender

- User Context (identity · location · chats)
- 4-channel recall
 - Similar People
 - Nearby People
 - Connectors
 - Collaborative Signals
- Privacy & Safety Filter
- Rank
- Recommendation
- Actions: Nudge · Intro · JoinActivity · Cohost

Strong skeleton. v2 keeps this · extends along 4 axes →

v2 · WHAT WE ADD

Heterogeneous · multi-trigger · time-aware

- 1 HETEROGENEOUS TARGETS**
Not just peer→peer · 6 target types: neighbor · meet · item · tip · event · activity
- 2 3 TRIGGER SOURCES**
L1 explicit query · L3a flow-driven · L3b content-driven · same engine · 3 callers
- 3 TIME-AWARE**
Every recommendation has TTL · decay scoring · expiry · the swap card rots after 14d
- 4 BIDIRECTIONAL TRUST**
Per-pair not per-target · A→B tier ≠ B→A tier · enforced at filter, not data layer

+ All 5 gaps Yunchao flagged (impression storage · community ranking · reusable RPC · feedback loop · reason builder) — addressed in v2.

Three compounding loops · the moat math

every loop multiplies defensibility · 2 years of usage = unreplicable graph

1

SIGNAL FLYWHEEL

captured signals → emerging categories → roadmap candidates

mom asks for X
(in-scope or out-of-scope · doesn't matter)
→ signal logged
→ Signal Aggregator clusters
→ density crosses threshold
→ roadmap candidate emerges
→ next sprint we ship the capability
→ next mom asking for X gets fulfillment

more usage = more capabilities = fewer captures = more fulfillment

2

IDENTITY FLYWHEEL

claims refine matches refine engagement refine claims

mom describes herself
→ 7-dim claims extracted
→ embedding refined
→ matches improve
→ mom engages more
→ mom mentions more about herself
→ more claims surface
→ richer identity → better matches

richer identity → better matches → more conversations → richer identity

3

FEEDBACK FLYWHEEL

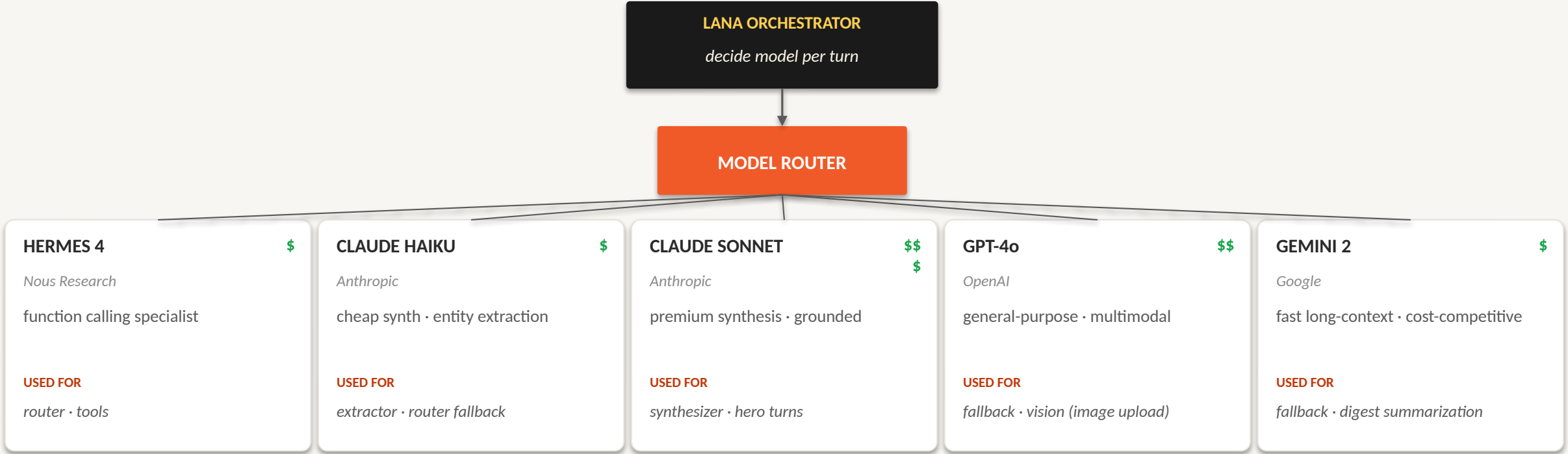
every recommendation surface is a labeled training example

recommendation shown
→ accepted / dismissed / ignored / converted
→ logged to recommendation_impressions
→ ranker retrains nightly
→ next recommendation better-ranked
→ L3b learns entity→capability associations
→ per-mom personalization emerges
→ ranker compounds with every mom-day

labeled data scales with usage · the moat compounds faster than any competitor's bootstrap

Model abstraction · cost-aware routing

vendor-independent · per-turn-type model selection · live cost dashboard · founder lock

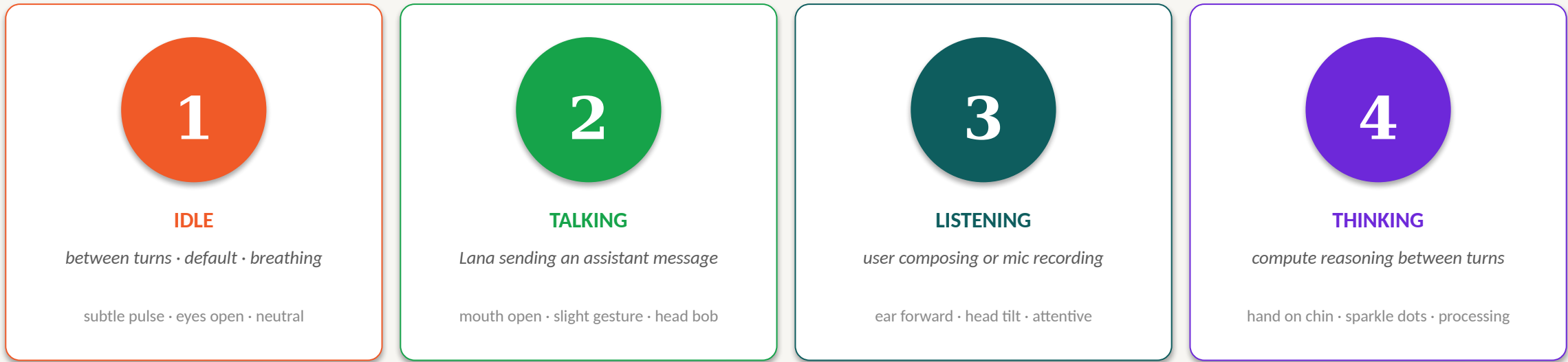


★ IP · COST-AWARE INFRASTRUCTURE · LANA CORP DASHBOARD (separate workstream)

Estimated marginal cost ~\$0.56/mom/month at 1k scale. Per-mom budget enforced via routing rules. Dashboard surfaces live cost-per-active-user, model-mix, and override controls. Built as a separate niche product (Lana Corp · founder-led).

Lana the character · 4 states · voice-first · multi-modal

voice is primary · text is secondary · image is for listings · video / AR in v1.0+



MODALITIES · primary → secondary → tertiary	
VOICE	primary input · mom drives kids · hands-free · the bell IS the action
TEXT	secondary · when mom can type · keyboard always available
IMAGE	for listings · mom uploads photo → Lana drafts the swap entry from the image
VIDEO / AR (v1.0+)	block walks · in-person meet AR overlays · long arc

Phased rollout · v0.1 → v2.0

AI-augmented dev velocity = we are aggressive on dates · founder lock

v0.1 CLOSED BETA	v0.2 PUBLIC OPEN	v0.3 MULTI-PERSONA	v1.0 MULTI-METRO	v2.0 PLATFORM
June 18-22 · NEXT WEEK 20 seed moms · Lake Nona East · PWA • 23 Layer 1 intents shipped • 14 Layer 2 categories captured (no UI yet) • 5 Concierge actions (nudge · intro · join · cohost · capture) • Block-Log preview in C-13 RADAR • Push (APNS) + SMS (Twilio) + Email (Resend) • 1-state Lana mascot (4-pose SVGs in design)	August 2026 200 moms · 5 ZIPs · native iOS + Android • Signal Aggregator dashboard LIVE • Heterogeneous recommendations (6 target types) • Layer 3a circumstantial in production • Feedback impression loop active • 4-state Lana mascot in production • Brand voice locked: "May I, {name}..." pattern • Mirroring strategy (Lana matches user's tone)	Q4 2026 2,000 moms · Orlando metro · 2nd persona • Layer 3b inferential in production • Per-mom personalization layer • Proactive Lana (1.5-2 nudges/day) • Second persona unlocked (runners or expats) • Multi-language production (EN/PT/ES)	Q2 2027 50K moms · top-10 US metros • Multi-persona platform live • Lana skill marketplace (community capabilities) • Monetization: concierge-mediated commerce • Lana Corp dashboard live • Federated ranker per ZIP	2028 500K users · international (LATAM + UK + AU) • Full agentic platform • Per-block federated learning • Multi-modal (voice + image + video) • Layer 3c predictive (foresight) • Lana-as-OS · communities deploy variants

Brand voice · frequency · the Lana personality lock

founder decisions locked · 2026-06-12

PROACTIVE VOICE PATTERN

"May I, {name}..."

deferential · uses mom's name · clear consent ask

"May I, Maria — moms in your block with karate kids meet Saturdays. Want me to find you a fit?"

Falls back to "by the way..." if proactive turn isn't a suggestion (e.g., reminder, check-in).

FREQUENCY CAP · proactive Lana

1.5 - 2 / day

morning + evening rhythm · not aggressive · not quiet

Quiet hours: 9pm-7am (use ZIP→TZ derivation)

Max 5 / week per mom · hard reset on dismissal of a category

24h cool-down on same-capability after dismissal

OPT-OUT MODEL · default ON · user adjusts via settings

No explicit consent dialog at signup. Lana introduces herself · proactive is default.

Settings affordance · LLM-style talkativeness slider:

- "Talk less" → 0.5 / day (light)
- "Default" → 1.5-2 / day (balanced)
- "Talk more" → 3 / day (engaged)

v0.3+: in-conversation natural commands · "Lana, you're too talkative."

MIRRORING STRATEGY · Lana matches the user

If mom writes in short bursts → Lana shortens.

If mom writes long paragraphs → Lana expands.

If mom uses voice → Lana leans conversational.

If mom uses formal language → Lana matches register.

Implemented in synthesizer prompt + voice rules.

Future: per-mom learned style profile (v0.3+).

Safety architecture · what nobody else builds carefully

minimum infrastructure · not overblown · scales with cohort size · new tables for emotional + adversarial data

IDENTITY VERIFICATION

5-stage ladder · in production

- Stranger → Nudge: phone verified (Supabase OTP)
- Nudge → Acquaintance: mutual nudge accepted
- Acquaintance → Direct: 1st co-attended meet
- Direct → IRL: ID-verified via third party (Persona/Stripe Identity)
- ↳ trigger: "want to meet in person" → upload ID → human review
- Founder lock: identity verification day one · minimum viable

EMOTIONAL DISTRESS · NEW TABLE

the 'alarming table' (founder lock)

- Separate from out_of_scope_inquiry
- Schema: distress_signals(user_id, turn_text, sentiment, urgency, escalation_flag, reviewer_notes, resolved_at)
- Detected by sentiment classifier on every turn
- Triggers: suicidal ideation · abusive relationship · ED indicators
- · acute kid-illness panic · custody/legal distress
- Escalation: founder daily review · crisis line referral copy

PROFANITY + SEX · NEW TABLES

separated for moderation review

- Schema: moderation_flags(user_id, turn_text, category, severity)
- Categories: profanity · sexual_content · harassment · spam · scam
- Auto-redaction in nudges before send to peer
- Repeat offenders: rate-limit + manual review
- Day Zero: simple deny-list + LLM classifier
- v0.3+: trained moderation model (Llama Guard)

Team · who owns what · sync rhythm

sharp boundaries · doc-driven · weekly Monday sync · no scope creep

YUNCHAO

AI lead

OWNS

- model selection
- prompt engineering
- Layer 1 intent classifier
- Layer 3 architecture
- capability index
- ranker (heuristic → ML)
- evals (golden + adversarial)
- reason builder library

BOUNDARY

DON'T own: schema, UX, product spec

ASJID

Backend lead

OWNS

- Supabase schema + migrations
- lana-worker service
- API contracts
- notification dispatch (SMS+push+email)
- signal aggregator pipeline
- ranker host (deploy what Yunchao trains)
- model abstraction layer
- observability + cost dashboard

BOUNDARY

DON'T own: prompts, model choice, UX

ABDULLAH

Frontend lead

OWNS

- PWA + native mobile UI
- v0.2.10 mockup → production parity
- ui_intent switcher
- auth flow (Supabase OTP)
- 4-state Lana mascot integration
- compass-aggregator nav
- Block-Log surface
- trust-tier UI gating

BOUNDARY

DON'T own: AI logic, schema

TOMMASO

Founder

OWNS

- product spec · brand
- Lana voice / tone
- GTM · cohort selection
- founding moms recruitment
- investor narrative
- safety call (final)
- Lana Corp dashboard (separate)
- Day Zero ops

BOUNDARY

Final say: product · safety · brand

WEEKLY CADENCE · Monday 30-min sync

agenda: blockers · new schema asks · production observations · 5-min decisions only · longer items → async doc

Reference docs + founder decisions locked

read these in order · all live in [R&D] TagAlong/ · keep Drive in sync

READING ORDER · v1 canon · 2026-06-12

01	LANA_BLUEPRINT_v1.md	the master · everyone reads first
02	LANA_LATENT_INTENT_ENGINE_v1.md	the moat layer · Yunchao + Asjid
03	LANA_RECOMMENDATION_ENGINE_v2.md	response to Yunchao's v1 · heterogeneous
04	LANA_INTENTS_AND_ROUTING_v1.md	23 in-scope + 14 fallback catalog
05	YUNCHAO_AI_HANDOFF_v1.md	his operating brief
06	ASJID_BACKEND_HANDOFF_v1.md	his operating brief + DDL
07	ABDULLAH_FRONTEND_HANDOFF_v1.md	his operating brief + API contract
08	BLOCK_LOG_SPEC_v1.md	the missing match-log surface
09	LANA_STATE_MODEL_v1.md	4-state mascot + animations
10	v0.2.8_PLAN.md	transcript synthesis + Day Zero plan
11-15	DRIVE canon (kept)	Architecture · Modus · Routing · Out-of-scope · Aggregator
16	LANA_UNIFIED_DISCOVERY_FRONTEND.md	GitHub · the shipped backend contract
WAVE 2	Eval · Safety · Observability · Knowledge Graph	this week
WAVE 3	Unit Economics · Privacy · Multimodal	next week

★ DECISIONS LOCKED · 2026-06-12

Voice	"May I, {name}..." pattern
Frequency	1.5-2/day · morning + evening
Opt model	default ON · settings adjust talkativeness
Mirroring	Lana matches user's tone/length
Model stack	abstraction layer · swappable · cost-aware routing
Multi-modal	voice primary · text 2nd · image for listings
Cohort go-live	next week · 20 moms · PWA · be aggressive on dates
Distress data	separate alarming table · founder daily review
Safety / ID	5-stage ladder + 3rd party ID for IRL meets
Dashboard	Lana Corp · separate workstream · founder-led

OPEN FOR NEXT SYNC · 3 critical

Per-mom \$ budget	ceiling drives model selection
Day Zero cohort	Lake Nona East vs Winter Garden vs both
Lana Corp scope	in or out of this Cowork session

What to do · this weekend

ship Monday with alignment · the team kicks off v0.2 architecture

YUNCHAO

AI

Read LANA_BLUEPRINT §2-5

Read LANA_LATENT_INTENT_ENGINE end-to-end

Read LANA_RECOMMENDATION_ENGINE v2 (it's about your v1)

Read YUNCHAO_AI_HANDOFF · answer §6 decision points D1-D5

By Monday: lock the 23-intent prompt + seed capability_index

ASJID

Backend

Read LANA_BLUEPRINT §3-5 · §10.4-10.5

Read ASJID_BACKEND_HANDOFF end-to-end

Review the 8-table migration plan · push back

Add Resend API key to Vercel env (provided · don't commit)

By Monday: tier-1 migration applied to dev

ABDULLAH

Frontend

Read LANA_BLUEPRINT §3 · §6

Read ABDULLAH_FRONTEND_HANDOFF + LANA_UNIFIED_DISCOVERY_FRONTEND

Audit v0.2.10 mockup vs your component library

Identify the ~10 highest-effort frames to port first

By Monday: production-port branch open · 3 frames done

TOMMASO · founder

Forward this deck + 7 docs to Yunchao · Asjid · Abdullah Sunday night. Drop in #engineering channel. Schedule the Monday 30-min sync.